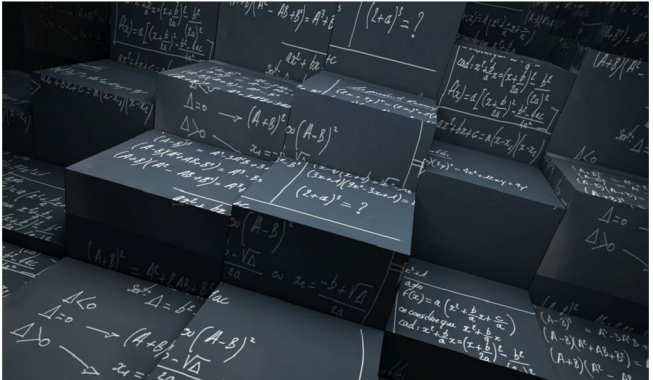


An Introduction to SageMath

Sage is wonderful open source software as it covers almost all areas of mathematics, with varying levels of sophistication. It has all the built-in objects and functions for expressing mathematical concepts and calculations. Besides a command line and programming library interface, it also has a dynamic self-hosted website.



Many of us are unaware of some of the best software that helps make our tasks easier. This article intends to introduce SageMath to open source users interested in mathematics, and also to beginners who wish to explore open source projects and contribute to them.

Here goes the mission statement of SageMath: “Create a viable, free, open source alternative to Magma, Maple, Mathematica and Matlab.”

SageMath is a free open source mathematics software system written in Python and Cython. It integrates over 90 open source mathematics software packages. William A. Stein is the lead developer of SageMath. It has a Web based graphical interface, and the user interface is a notebook in a Web browser or the command line. SageMath can be used in various areas of mathematics like basic algebra, linear algebra, numerical computation, graph theory, cryptography and much more. And since it is used in areas like Calculus, linear algebra, abstract algebra, etc, it is helpful in preparing curriculum for undergraduate students. Many standard textbooks for undergraduates include solutions to their exercises using Sage, which we can refer to in the following link: <http://www.sagemath.org/library-publications.html#books>

Installation

The first version of Sage was released on February 24, 2015.
The present stable version is Sage 6.8.

To install Sage on Linux (Ubuntu/Debian), your system should have the C/C++ compiler, *make*, *m4*, Perl, *ar* and *ranlib* (which can be obtained as part of GNU *binutils*) and *tar* (Figure 1).



Figure 1: Installation prerequisites